

NEXUS COMMAND

TRIAL RUN BLUEPRINT — BACKEND + FRONTEND ALIGNMENT BRIEF

31 May 2026

For: Base44 Builder AI · From: Simpee · Parallel with backend changes already deployed

Fix 4 issues · Test 5 checkpoints · Prove the core loop

PURPOSE: This brief tells the Base44 builder exactly what is already built on the backend so the frontend can be wired to match precisely. No guessing. No misalignment. Every function, every field name, every expected payload and response shape is documented here. The builder only needs to wire the frontend to what already exists.

WHAT IS ALREADY BUILT — Backend State as of Today

The builder does NOT need to create any of these. They exist and are tested. The frontend just needs to call them correctly.

Item	Name / ID	Status	Notes
Function	aiCommandCentre	LIVE	Responds to instruction. Returns analysis, solution, code, builder_instruction. Currently in lite m
Function	consultCopilot	LIVE — field mismatch	Expects { question, context } NOT { code, context }. Frontend must send 'question' not 'code'. F
Function	checkConnections	LIVE	Pings all AI connectors. Returns status per connection.
Function	syncToOneDrive	LIVE	Saves session to OneDrive. Called after deploy.
Entity	SChatMessage	EXISTS · 1 record	Fields: sender, sender_type, role, message, timestamp, session_id, read
Entity	TestLog	EXISTS · 1 record	Fields: test_name, status, result, tested_at, fixed, validator
Entity	AIConnector	EXISTS · 8 records	All 8 READY. Fields: model_name, intro_message, voice_style, handoff_to, status, provider, m
Entity	azureConnectorStub	EXISTS · 1 record	Fields: service_name, status, last_tested, notes
Entity	SandboxProject	EXISTS · 0 records	Fields: name, description, type, status, created_by, checkpoint_status, validator_approved, not
Entity	ProjectFile	EXISTS · 1 record	Fields: project_id, filename, content, language, version, notes

4 FRONTEND FIXES — Exact Changes Required

These are the only changes needed. Do not rebuild the layout. Do not change entity schemas. Wire the existing UI to the existing backend correctly.

FIX 1 Wire chat input → aiCommandCentre correctly

Function URL: POST /api/run/aiCommandCentre Send this exact payload: { "instruction": "[user's typed message]", "posted_by": "Kieran" } On success, display in centre panel: Card header: model name from response.model Section 1 – ANALYSIS: response.analysis Section 2 – SOLUTION: response.solution Section 3 – CODE: response.code (monospace block, copy button) Section 4 – BUILDER INSTRUCTION: response.builder_instruction (amber bg) Show loading state: 'Simpee is working...' with animated dots while awaiting response.

FIX 2 Fix consultCopilot payload — change 'code' to 'question'

Function URL: POST /api/run/consultCopilot WRONG (current): { "code": "...", "context": "..." } CORRECT (fix to): { "question": "...", "context": "..." } Where to get the values: question = response.code from the last aiCommandCentre response context = response.builder_instruction from the same response The [RUN VALIDATION] button in the Gatekeeper Panel fires this call. On response, update the 4 Gatekeeper check rows: Syntax · Logic · Security · Azure Each row: show green tick if passed, red flag if issue found. Write 1 TestLog record per check with validator='Copilot'.

FIX 3 Auto-write TestLog after every aiCommandCentre response

After every successful aiCommandCentre response, write to TestLog entity: `POST /api/entities/TestLog { "test_name": "Trial Run - " + instruction.slice(0,40), "status": "pass", "result": "aiCommandCentre responded. Model: " + response.model, "tested_at": new Date().toISOString(), "fixed": false, "validator": "Simpee" }` This auto-logs checkpoint 3 (aiCommandCentre responds) for every session.

FIX 4 Add Sandbox Project card + [SAVE CODE TO SANDBOX] button

In the right panel, below the Deploy Log section, add a slim Sandbox card: 1. On load: fetch SandboxProject entity – show the first record (trial project). Display: name, status badge (DRAFT=grey / TESTING=amber / APPROVED=green), checkpoint_status 2. Add [SAVE CODE TO SANDBOX] button (Azure #0078d4 bg). On click: write to ProjectFile entity: `{ "project_id": [sandbox project id], "filename": "response_" + Date.now() + ".ts", "content": [response.code from last aiCommandCentre response], "language": "typescript", "version": "1.0", "notes": "Saved from trial run" }` Show success toast: 'Code saved to Sandbox' 3. Show file count below: 'X files saved to this project'

TRIAL RUN TEST CHECKLIST — 5 Checkpoints to Pass

Kieran will run these tests after the builder delivers. Simpee will verify each one against the entity records.

#	Test	Action	Expected Result	Simpee Verifies By
1	Chat → AI response	Type any message in chat input	Response card appears with ANALYSIS, SOLUTION, CODE Message entity loading dots shown while working	ChatMessage entity loading dots shown
2	TestLog auto-write	Same as test 1 — no extra test log needed	TestLog gets new record with validator=Simpee	TestLog response appears
3	Gatekeeper validation	After a response, click [RUN VALIDATION] in right panel	4 Gatekeeper check rows (Syntax / Logic / Security / Azure) with green tick or red flag	Each shows validator=Copilot. 4 TestLog records with validator='Copilot'
4	Save to Sandbox	Click [SAVE CODE TO SANDBOX] button	SandboxProject appears. ProjectFile entity has 1 record	Read ProjectFile entity. SandboxProject_id matches
5	Full loop confirmed	Run tests 1-4 in sequence	All 4 entity writes confirmed: SChatMessage (4 tests) + STestLog (8 checks) + STestLog (Copilot) + ProjectFile (1)	

DO NOT TOUCH — Protected Items

Do NOT change	Reason
aiCommandCentre.ts backend code	Already tested and working. Only frontend payload needs fixing.
consultCopilot.ts backend code	Field mismatch is a frontend issue only — the function expects 'question' and that is correct.
Entity schemas	All schemas are correct and aligned. Do not modify.
Left panel AI selector layout	Not part of this trial run scope.
Any other pages or components	Scoped fix only — centre panel + right panel + chat input only.

BUILDER PREFIX — Paste This into Nexus Command Builder

"Fix 4 frontend wiring issues in the Nexus Command app (CommandAIHub.jsx). Do NOT change any backend functions or entity schemas — they are already correct. FIX 1 — Chat input to aiCommandCentre: Send `POST /api/run/aiCommandCentre` with `{ instruction: userMessage, posted_by: 'Kieran' }`. Show loading dots while waiting. On response, render 4 sections in centre panel: ANALYSIS · SOLUTION · CODE (monospace + copy button) · BUILDER INSTRUCTION (amber bg). Label card with response.model. FIX 2 — consultCopilot field name: The [RUN VALIDATION] button must send `{ question: lastCode, context: lastBuilderInstruction }` NOT `{ code: ... }`. One word change. On response update 4 Gatekeeper check rows (Syntax / Logic / Security / Azure) with green tick or red flag. Write 4 TestLog records with validator='Copilot'. FIX 3 — Auto TestLog write: After every successful aiCommandCentre response, write to TestLog: `{ test_name: 'Trial Run - '+instruction.slice(0,40), status: 'pass', result: 'Model: '+response.model, tested_at: new Date().toISOString(), fixed: false, validator: 'Simpee' }`. FIX 4 — Sandbox card in right panel: Below Deploy Log, add a slim card. Fetch SandboxProject entity on load, show first record (name, status badge, checkpoint_status). Add [SAVE CODE TO SANDBOX] button (Azure #0078d4). On click: write to ProjectFile entity with `project_id, filename=response_[timestamp].ts, content=lastCode, language=typescript, version=1.0`. Show success toast. Design system: bg=#e8e6fe, accent=#5e50fb, Azure=#0078d4, Exo 2 headlines, Montserrat body. No emoji."